

SMART TRANSIT NETWORK SYSTEM

Topic 11 | Group 803

Base Network: 20.20.0.0/16

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1. Project Overview

The Smart Transit Network System connects six operational centres of a city transit authority into one routed internetwork. The supplied specification requires VLSM, five different static-route forms, RIPv2, DHCP, DNS, web and email services, and a backup path.

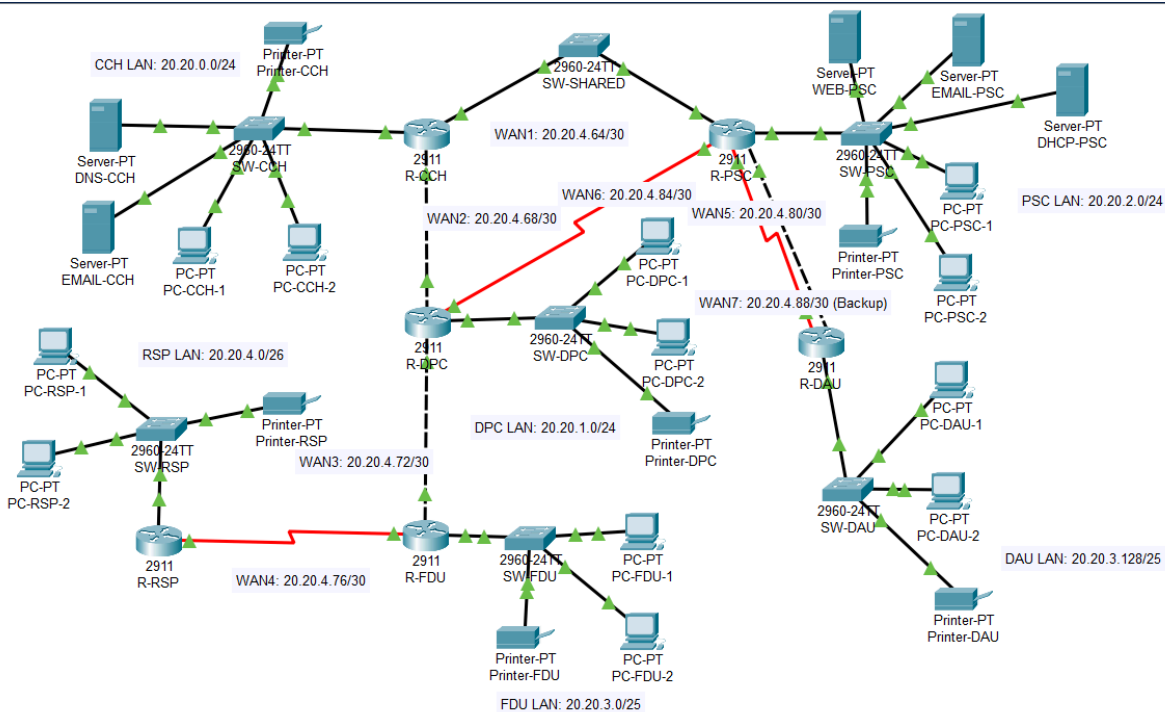
1.1 Centres and Requirements

Centre	Devices required	Host addressing	Services
CCH	200	Static only	DNS, Email
DPC	150	Router-based DHCP	—
PSC	135	Dedicated DHCP server	Web, Email, DHCP
FDU	120	Router-based DHCP	—
DAU	85	Static only	—
RSP	50	Static only	—

1.2 Link Inventory

Link	Connection	Subnet	Type
WAN1	CCH–PSC via shared switch	20.20.4.64/30	Gigabit/shared Layer 2
WAN2	CCH–DPC	20.20.4.68/30	Gigabit
WAN3	DPC–FDU	20.20.4.72/30	Gigabit
WAN4	FDU–RSP	20.20.4.76/30	Serial
WAN5	PSC–DAU	20.20.4.80/30	Gigabit
WAN6	DPC–PSC backup	20.20.4.84/30	Backup serial
WAN7	DAU–PSC backup	20.20.4.88/30	Backup serial

1.3 Topology Diagram



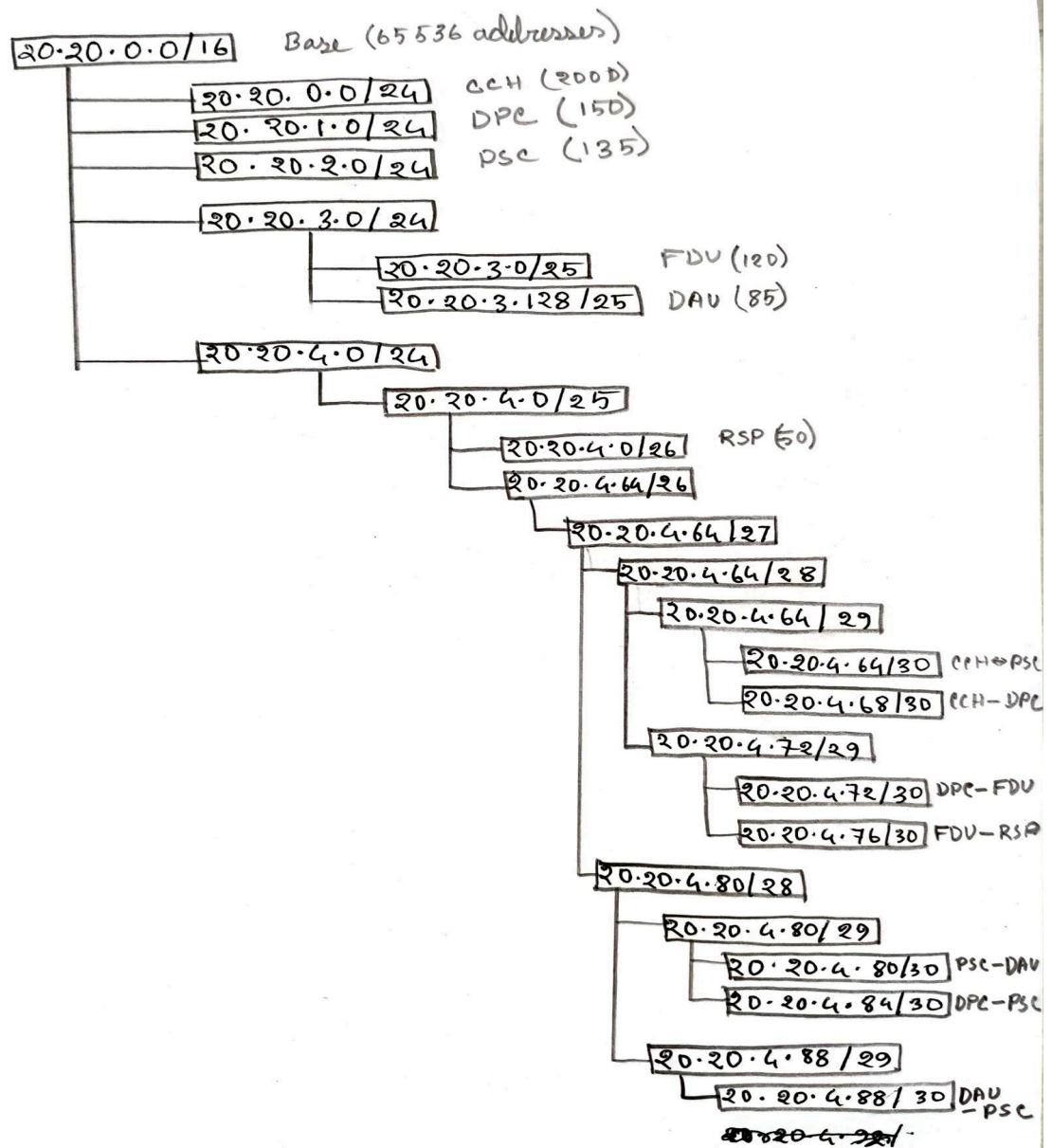
2. Base Network Derivation

This is a 4-member group, so the base network was derived from the sum of the last three digits of the fourth member's student ID (Samiul Mahmud, 22299398): $3 + 9 + 8 = 20$. This gives the base network 20.20.0.0/16 (subnet mask 255.255.0.0).

3. VLSM Addressing Plan, VLSM Tree and IP ADDRESS Table

Network	Subnet mask	Usable range	Gateway / endpoints
20.20.0.0/24 (CCH)	255.255.255.0	20.20.0.1–.254	20.20.0.1
20.20.1.0/24 (DPC)	255.255.255.0	20.20.1.1–.254	20.20.1.1
20.20.2.0/24 (PSC)	255.255.255.0	20.20.2.1–.254	20.20.2.1
20.20.3.0/25 (FDU)	255.255.255.128	20.20.3.1–.126	20.20.3.1
20.20.3.128/25 (DAU)	255.255.255.128	20.20.3.129–.254	20.20.3.129
20.20.4.0/26 (RSP)	255.255.255.192	20.20.4.1–.62	20.20.4.1
20.20.4.64/30 (WAN1)	255.255.255.252	20.20.4.65–.66	CCH .65 / PSC .66
20.20.4.68/30 (WAN2)	255.255.255.252	20.20.4.69–.70	CCH .69 / DPC .70
20.20.4.72/30 (WAN3)	255.255.255.252	20.20.4.73–.74	DPC .73 / FDU .74
20.20.4.76/30 (WAN4)	255.255.255.252	20.20.4.77–.78	FDU .77 / RSP .78
20.20.4.80/30 (WAN5)	255.255.255.252	20.20.4.81–.82	PSC .81 / DAU .82
20.20.4.84/30 (WAN6)	255.255.255.252	20.20.4.85–.86	DPC .85 / PSC .86
20.20.4.88/30 (WAN7)	255.255.255.252	20.20.4.89–.90	DAU .89 / PSC .90

VLSM TREE:



There are 12 allocated subnets, zero overlap, zero gaps inside the allocated range, and 1,112 addresses consumed from the /16. The DAU–PSC backup link (WAN7, 20.20.4.88/30) was added afterward as a 13th subnet, consuming 4 further addresses immediately after the WAN6 block with no overlap or gap.

4. Router Interface Addressing

Router	Interface	IP	Mask	Role
R-CCH	G0/0	20.20.0.1	255.255.255.0	CCH LAN
R-CCH	G0/1	20.20.4.65	255.255.255.252	WAN1 to PSC
R-CCH	G0/2	20.20.4.69	255.255.255.252	WAN2 to DPC
R-DPC	G0/0	20.20.1.1	255.255.255.0	DPC LAN
R-DPC	G0/1	20.20.4.70	255.255.255.252	WAN2 to CCH
R-DPC	G0/2	20.20.4.73	255.255.255.252	WAN3 to FDU
R-DPC	S0/0/1	20.20.4.85	255.255.255.252	WAN6 to PSC
R-FDU	G0/0	20.20.3.1	255.255.255.128	FDU LAN
R-FDU	G0/1	20.20.4.74	255.255.255.252	WAN3 to DPC
R-FDU	S0/3/0	20.20.4.77	255.255.255.252	WAN4 to RSP
R-PSC	G0/0	20.20.2.1	255.255.255.0	PSC LAN
R-PSC	G0/1	20.20.4.66	255.255.255.252	WAN1 to CCH
R-PSC	G0/2	20.20.4.81	255.255.255.252	WAN5 to DAU
R-PSC	S0/3/0	20.20.4.86	255.255.255.252	WAN6 to DPC
R-DAU	G0/0	20.20.3.129	255.255.255.128	DAU LAN
R-DAU	G0/1	20.20.4.82	255.255.255.252	WAN5 to PSC
R-DAU	S0/0/0	20.20.4.89	255.255.255.252	WAN7 (backup) to PSC
R-PSC	S0/2/0	20.20.4.90	255.255.255.252	WAN7 (backup) to DAU

R-RSP	G0/0	20.20.4.1	255.255.255.192	RSP LAN
R-RSP	S0/3/0	20.20.4.78	255.255.255.252	WAN4 to FDU

5. Host, Server and Switch Addressing

Site	Device	Address / allocation	Gateway	Method
CCH	SW-CCH	20.20.0.2	20.20.0.1	Static
CCH	DNS Server	20.20.0.10	20.20.0.1	Static
CCH	Email Server	20.20.0.11	20.20.0.1	Static
CCH	PC-CCH-1 / 2	20.20.0.21 / .22	20.20.0.1	Static
CCH	Printer	20.20.0.30	20.20.0.1	Static
DPC	SW-DPC	20.20.1.2	20.20.1.1	Static
DPC	PC-DPC-1 / 2	DHCP from .21 upward	20.20.1.1	Router DHCP
DPC	Printer	20.20.1.10	20.20.1.1	Static/excluded
PSC	SW-PSC	20.20.2.2	20.20.2.1	Static
PSC	Web Server	20.20.2.10	20.20.2.1	Static
PSC	Email Server	20.20.2.11	20.20.2.1	Static
PSC	DHCP Server	20.20.2.12	20.20.2.1	Static
PSC	PC-PSC-1 / 2	DHCP from .100 upward	20.20.2.1	Dedicated server
PSC	Printer	20.20.2.30	20.20.2.1	Static/excluded
FDU	SW-FDU	20.20.3.2	20.20.3.1	Static
FDU	PC-FDU-1 / 2	DHCP from .21 upward	20.20.3.1	Router DHCP
FDU	Printer	20.20.3.10	20.20.3.1	Static/excluded
DAU	SW-DAU	20.20.3.130	20.20.3.129	Static
DAU	PC-DAU-1 / 2	20.20.3.140 / .141	20.20.3.129	Static
DAU	Printer	20.20.3.148	20.20.3.129	Static
RSP	SW-RSP	20.20.4.2	20.20.4.1	Static

RSP	PC-RSP-1 / 2	20.20.4.11 / .12	20.20.4.1	Static
RSP	Printer	20.20.4.20	20.20.4.1	Static

6. Routing Design

Route type	Router	Requirement
Next-hop static	R-CCH	20.20.3.0/25 via 20.20.4.70 (DPC)
Exit-interface static	R-RSP	20.20.0.0/24 out Serial0/3/0
Recursive static	R-DAU	20.20.4.0/26 via 20.20.4.81 (PSC), then PSC recursively reaches RSP through CCH-DPC-FDU
Floating static	R-DPC	20.20.2.0/24 via 20.20.4.86 at AD 130
Default static	R-RSP	0.0.0.0/0 via 20.20.4.77; only RSP has a default route
Floating static (backup)	R-DAU	20.20.4.0/26 via 20.20.4.90 (WAN7, PSC) at AD 130 — backup for the primary recursive route via 20.20.4.81
Floating static (backup)	R-PSC	20.20.3.128/25 via 20.20.4.89 (WAN7, DAU) at AD 130 — backup for the primary route to the DAU LAN

6.1 RIPv2 and Redistribution

RIPv2 runs on CCH, DPC, FDU and PSC with version 2 and no auto-summary. DAU and RSP are static-only. CCH redistributes its static routes into RIP using metric 5. The DPC–PSC backup serial link is passive in RIP so the floating static remains dormant during normal operation.

6.2 DAU–PSC Backup Link (WAN7)

A second backup path was added between DAU and PSC (WAN7, 20.20.4.88/30, on Serial0/0/0 at R-DAU and Serial0/2/0 at R-PSC) to provide redundancy for the DAU–PSC primary link, mirroring the existing DPC–PSC backup design. The backup is bidirectional: R-DAU carries a floating static route to 20.20.4.0/26 via 20.20.4.90 at administrative distance 130 (backing up its primary recursive route via 20.20.4.81), and R-PSC carries a floating static route to the DAU LAN, 20.20.3.128/25, via 20.20.4.89, also at AD 130 (backing up its primary route to DAU). Both floating routes stay dormant behind their respective AD-1 routes until the primary DAU–PSC link fails. Since DAU and RSP do not run RIP, no passive-interface change is required at the DAU end for this link.

7. DHCP and Server Configuration Summary

Service	Location	Configuration
DPC DHCP	R-DPC	DPC-POOL; 20.20.1.21–20.20.1.254; gateway 20.20.1.1; DNS 20.20.0.10
FDU DHCP	R-FDU	FDU-POOL; 20.20.3.21–20.20.3.126; gateway 20.20.3.1; DNS 20.20.0.10
PSC DHCP	20.20.2.12	PSC-POOL; 20.20.2.100–20.20.2.199; gateway 20.20.2.1
DNS	20.20.0.10	A record www.transitline.local -> 20.20.2.10
Web	20.20.2.10	HTTP service for www.transitline.local
Email	20.20.0.11 / 20.20.2.11	CCH and PSC email services

8. Assumptions and Design Decisions

- The redistribution metric for CCH's static routes into RIP was deliberately set to 5 (not the default 1), so that redistributed routes remain clearly distinguishable in the routing table from routes learned natively through RIP, without affecting route selection.
- passive-interface was applied on both ends of the DPC-PSC backup serial link (WAN6). Without this, RIP would learn a route over the backup link and always prefer it (administrative distance 120) over the floating static route (administrative distance 130), so the floating route would never activate as required.
- The email domain names (cch.com and psc.com) were chosen freely, since the project brief only specifies a fixed domain for the web server (www.transitline.local); DNS and web service names follow the brief exactly.
- The failure scenario "If the direct link between DAU and PSC fails, DAU must still reach RSP using the recursive static route (DAU → PSC → CCH → DPC → FDU → RSP)" cannot be physically demonstrated as literally worded, because DAU has only one physical uplink to the network and that uplink is PSC itself - the very link the stated path depends on as its first hop. We instead verified the substantive requirement: PSC holds a valid recursive static route to RSP (via CCH→ DPC → FDU), and DAU successfully reaches RSP end-to-end through this exact path under normal operation (see traceroute in the verification screenshots). Taking the DAU-PSC link down was also tested and, as expected from the topology, isolates DAU completely, which confirms there is no alternate path.

9. Configuration Commands

R-CCH

```
enable
configure terminal
hostname R-CCH
no ip domain-lookup
ip domain-name transitline.local
enable secret class
line console 0
password cisco
login
logging synchronous
exec-timeout 0 0
line vty 0 4
password cisco
login
transport input all
service password-encryption
banner motd # R-CCH - Group 803 Smart Transit Network. Authorised access only. #

interface GigabitEthernet0/0
ip address 20.20.0.1 255.255.255.0
interface GigabitEthernet0/1
ip address 20.20.4.65 255.255.255.252
interface GigabitEthernet0/2
ip address 20.20.4.69 255.255.255.252

router rip
version 2
redistribute static metric 5
network 20.0.0.0
no auto-summary

ip route 20.20.1.0 255.255.255.0 20.20.4.70
ip route 20.20.2.0 255.255.255.0 20.20.4.66
ip route 20.20.3.0 255.255.255.128 20.20.4.70
ip route 20.20.3.128 255.255.255.128 20.20.4.66
ip route 20.20.4.0 255.255.255.192 20.20.4.70

end
write memory
```

R-DPC

```
enable
configure terminal
hostname R-DPC
no ip domain-lookup
ip domain-name transitline.local
enable secret class
line console 0
  password cisco
  login
logging synchronous
exec-timeout 0 0
line vty 0 4
  password cisco
  login
  transport input all
service password-encryption
banner motd # R-DPC - Group 803 Smart Transit Network. Authorised access only. #

interface GigabitEthernet0/0
  ip address 20.20.1.1 255.255.255.0
interface GigabitEthernet0/1
  ip address 20.20.4.70 255.255.255.252
interface GigabitEthernet0/2
  ip address 20.20.4.73 255.255.255.252
interface Serial0/0/1
  ip address 20.20.4.85 255.255.255.252
  clock rate 64000

ip dhcp excluded-address 20.20.1.1 20.20.1.20
ip dhcp pool DPC_POOL
  network 20.20.1.0 255.255.255.0
  default-router 20.20.1.1
  dns-server 20.20.0.10
  domain-name transitline.local

router rip
  version 2
  passive-interface Serial0/0/1
  network 20.0.0.0
  no auto-summary

ip route 20.20.4.0 255.255.255.192 20.20.4.74
```

```
ip route 20.20.2.0 255.255.255.0 20.20.4.86 130

end

write memory
```

R-FDU

```
enable
configure terminal
hostname R-FDU
no ip domain-lookup
ip domain-name transitline.local
enable secret class
line console 0
password cisco
login
logging synchronous
exec-timeout 0 0
line vty 0 4
password cisco
login
transport input all
service password-encryption
banner motd # R-FDU - Group 803 Smart Transit Network. Authorised access only. #

interface GigabitEthernet0/0
ip address 20.20.3.1 255.255.255.128
interface GigabitEthernet0/1
ip address 20.20.4.74 255.255.255.252
interface Serial0/3/0
ip address 20.20.4.77 255.255.255.252
clock rate 64000

ip dhcp excluded-address 20.20.3.1 20.20.3.20
ip dhcp pool FDU_POOL
network 20.20.3.0 255.255.255.128
default-router 20.20.3.1
dns-server 20.20.0.10
domain-name transitline.local

router rip
version 2
network 20.0.0.0
no auto-summary
```

```
end
write memory
```

R-PSC

```
enable
configure terminal
hostname R-PSC
no ip domain-lookup
ip domain-name transitline.local
enable secret class
line console 0
password cisco
login
logging synchronous
exec-timeout 0 0
line vty 0 4
password cisco
login
transport input all
service password-encryption
banner motd # R-PSC - Group 803 Smart Transit Network. Authorised access only. #

interface GigabitEthernet0/0
ip address 20.20.2.1 255.255.255.0
interface GigabitEthernet0/1
ip address 20.20.4.66 255.255.255.252
interface GigabitEthernet0/2
ip address 20.20.4.81 255.255.255.252
interface Serial0/2/0
ip address 20.20.4.90 255.255.255.252
no shutdown
interface Serial0/3/0
ip address 20.20.4.86 255.255.255.252

router rip
version 2
passive-interface Serial0/3/0
network 20.0.0.0
no auto-summary

ip route 20.20.3.128 255.255.255.128 20.20.4.82
ip route 20.20.4.0 255.255.255.192 20.20.4.65
ip route 20.20.1.0 255.255.255.0 20.20.4.85 130
```

```
ip route 20.20.3.128 255.255.255.128 20.20.4.89 130

end
write memory
```

R-DAU

```
enable
configure terminal
hostname R-DAU
no ip domain-lookup
ip domain-name transitline.local
enable secret class
line console 0
  password cisco
  login
  logging synchronous
  exec-timeout 0 0
line vty 0 4
  password cisco
  login
  transport input all
service password-encryption
banner motd # R-DAU - Group 803 Smart Transit Network. Authorised access only. #
```

```
interface GigabitEthernet0/0
  ip address 20.20.3.129 255.255.255.128
interface GigabitEthernet0/1
  ip address 20.20.4.82 255.255.255.252
interface Serial0/0/0
  ip address 20.20.4.89 255.255.255.252
  no shutdown
```

```
ip route 20.20.4.0 255.255.255.192 20.20.4.81
ip route 20.20.0.0 255.255.255.0 20.20.4.81
ip route 20.20.1.0 255.255.255.0 20.20.4.81
ip route 20.20.2.0 255.255.255.0 20.20.4.81
ip route 20.20.3.0 255.255.255.128 20.20.4.81
ip route 20.20.4.64 255.255.255.252 20.20.4.81
ip route 20.20.4.68 255.255.255.252 20.20.4.81
ip route 20.20.4.72 255.255.255.252 20.20.4.81
ip route 20.20.4.76 255.255.255.252 20.20.4.81
ip route 20.20.4.84 255.255.255.252 20.20.4.81
```

```
ip route 20.20.4.0 255.255.255.192 20.20.4.90 130

end
write memory
```

R-RSP

```
enable
configure terminal
hostname R-RSP
no ip domain-lookup
ip domain-name transitline.local
enable secret class
line console 0
password cisco
login
logging synchronous
exec-timeout 0 0
line vty 0 4
password cisco
login
transport input all
service password-encryption
banner motd # R-RSP - Group 803 Smart Transit Network. Authorised access only. #

interface GigabitEthernet0/0
ip address 20.20.4.1 255.255.255.192
interface Serial0/3/0
ip address 20.20.4.78 255.255.255.252

ip route 20.20.0.0 255.255.255.0 Serial0/0/0
ip route 0.0.0.0 0.0.0.0 20.20.4.77

end
write memory
```

SW-CCH

```
enable
configure terminal
hostname SW-CCH
no ip domain-lookup
ip domain-name transitline.local
enable secret class
line console 0
password cisco
login
logging synchronous
exec-timeout 0 0
line vty 0 4
password cisco
login
transport input all
service password-encryption
banner motd # SW-CCH - Group 803 Smart Transit Network. Authorised access only. #

interface vlan 1
ip address 20.20.0.2 255.255.255.0
ip default-gateway 20.20.0.1

end
write memory
```

SW-DPC

```
enable
configure terminal
hostname SW-DPC
no ip domain-lookup
ip domain-name transitline.local
enable secret class
line console 0
password cisco
login
logging synchronous
exec-timeout 0 0
line vty 0 4
password cisco
login
transport input all
service password-encryption
banner motd # SW-DPC - Group 803 Smart Transit Network. Authorised access only. #
```

```
interface vlan 1
 ip address 20.20.1.2 255.255.255.0
 ip default-gateway 20.20.1.1
```

```
end
write memory
```

SW-PSC

```
enable
configure terminal
hostname SW-PSC
no ip domain-lookup
ip domain-name transitline.local
enable secret class
line console 0
 password cisco
 login
 logging synchronous
 exec-timeout 0 0
line vty 0 4
 password cisco
 login
 transport input all
service password-encryption
banner motd # SW-PSC - Group 803 Smart Transit Network. Authorised access only. #
```

```
interface vlan 1
 ip address 20.20.2.2 255.255.255.0
 ip default-gateway 20.20.2.1
```

```
end
write memory
```

SW-FDU

```
enable
configure terminal
hostname SW-FDU
no ip domain-lookup
ip domain-name transitline.local
enable secret class
```

```
line console 0
password cisco
login
logging synchronous
exec-timeout 0 0
line vty 0 4
password cisco
login
transport input all
service password-encryption
banner motd # SW-FDU - Group 803 Smart Transit Network. Authorised access only. #

interface vlan 1
ip address 20.20.3.2 255.255.255.128
ip default-gateway 20.20.3.1

end
write memory
```

SW-DAU

```
enable
configure terminal
hostname SW-DAU
no ip domain-lookup
ip domain-name transitline.local
enable secret class
line console 0
password cisco
login
logging synchronous
exec-timeout 0 0
line vty 0 4
password cisco
login
transport input all
service password-encryption
banner motd # SW-DAU - Group 803 Smart Transit Network. Authorised access only. #

interface vlan 1
ip address 20.20.3.130 255.255.255.128
ip default-gateway 20.20.3.129

end
```

write memory

SW-RSP

```
enable
configure terminal
hostname SW-RSP
no ip domain-lookup
ip domain-name transitline.local
enable secret class
line console 0
password cisco
login
logging synchronous
exec-timeout 0 0
line vty 0 4
password cisco
login
transport input all
service password-encryption
banner motd # SW-RSP - Group 803 Smart Transit Network. Authorised access only. #

interface vlan 1
ip address 20.20.4.2 255.255.255.192
ip default-gateway 20.20.4.1

end
write memory
```

SW-SHARED

```
enable
configure terminal
hostname SW-SHARED
no ip domain-lookup
ip domain-name transitline.local
enable secret class
line console 0
password cisco
login
logging synchronous
exec-timeout 0 0
line vty 0 4
```

```
password cisco
login
transport input all
service password-encryption
banner motd # SW-SHARED - Group 803 Smart Transit Network. Authorised access only. #

no ip address

end
write memory
```

10. Verification Evidence —Screenshots

A. Static Routes (5)

```
[OK]
R-CCH#show ip route | include 0.0.0.0
    20.0.0.0/8 is variably subnetted, 15 subnets, 5 masks
R-CCH#show run | include ip route
ip route 20.20.1.0 255.255.255.0 20.20.4.70
ip route 20.20.2.0 255.255.255.0 20.20.4.66
ip route 20.20.3.0 255.255.255.128 20.20.4.70
ip route 20.20.3.128 255.255.255.128 20.20.4.66
ip route 20.20.4.0 255.255.255.192 20.20.4.70
R-CCH#
```

```
-----
R-RSP#show ip route | include 0.0.0.0
Gateway of last resort is 20.20.4.77 to network 0.0.0.0
    20.0.0.0/8 is variably subnetted, 5 subnets, 4 masks
S*    0.0.0.0/0 [1/0] via 20.20.4.77
R-RSP#show run | include ip route
ip route 20.20.0.0 255.255.255.0 Serial0/3/0
ip route 0.0.0.0 0.0.0.0 20.20.4.77
```

```

R-DPC#show ip route | include 0.0.0.0
    20.0.0.0/8 is variably subnetted, 16 subnets, 5 masks
R-DPC#show run | include ip route
ip route 20.20.4.0 255.255.255.192 20.20.4.74
ip route 20.20.2.0 255.255.255.0 20.20.4.86 130
R-DPC#

R-PSC#show run | include ip route
ip route 20.20.3.128 255.255.255.128 20.20.4.82
ip route 20.20.4.0 255.255.255.192 20.20.4.65
ip route 20.20.1.0 255.255.255.0 20.20.4.85 130

R-DAU#show run | include ip route
ip route 20.20.4.0 255.255.255.192 20.20.4.81
ip route 20.20.0.0 255.255.255.0 20.20.4.81
ip route 20.20.1.0 255.255.255.0 20.20.4.81
ip route 20.20.2.0 255.255.255.0 20.20.4.81
ip route 20.20.3.0 255.255.255.128 20.20.4.81
ip route 20.20.4.64 255.255.255.252 20.20.4.81
ip route 20.20.4.68 255.255.255.252 20.20.4.81
ip route 20.20.4.72 255.255.255.252 20.20.4.81
ip route 20.20.4.76 255.255.255.252 20.20.4.81
ip route 20.20.4.84 255.255.255.252 20.20.4.81

```

Recursive chain

```

R-DAU#show ip route 20.20.4.0
Routing entry for 20.20.4.0/26
Known via "static", distance 1, metric 0
  Routing Descriptor Blocks:
    * 20.20.4.81
      Route metric is 0, traffic share count is 1

R-PSC#show ip route 20.20.4.0
Routing entry for 20.20.4.0/26
Known via "static", distance 1, metric 0
  Redistributing via rip
  Advertised by rip
  Routing Descriptor Blocks:
    * 20.20.4.65
      Route metric is 0, traffic share count is 1

```

```
R-CCH#show ip route 20.20.4.0
Routing entry for 20.20.4.0/26
Known via "static", distance 1, metric 0
  Redistributing via rip
  Advertised by rip
  Routing Descriptor Blocks:
    * 20.20.4.70
      Route metric is 0, traffic share count is 1
```

```
R-DPC#show ip route 20.20.4.0
Routing entry for 20.20.4.0/26
Known via "static", distance 1, metric 0
  Routing Descriptor Blocks:
    * 20.20.4.74
      Route metric is 0, traffic share count is 1
```

B. RIPv2 + Redistribution

```
R-FDU#show run | section router rip
router rip
  version 2
  network 20.0.0.0
  no auto-summary
```

```
R-DAU#show run | section router rip
R-DAU#
```

```
R-PSC#show run | section router rip
router rip
  version 2
  passive-interface Serial0/3/0
  network 20.0.0.0
  no auto-summary
R-PSC#
```

```
R-CCH#show run | section router rip
router rip
  version 2
  redistribute static metric 5
  network 20.0.0.0
  no auto-summary
R-CCH#
```

```
R-DPC#show run | section router rip
router rip
  version 2
  passive-interface Serial0/3/1
  network 20.0.0.0
  no auto-summary
n nnc#l
ip route 0.0.0.0 0.0.0.0 20.20.4.77
R-RSP#show run | section router rip
R-RSP#
```

Redistributed static routes (metric 5)

```

no auto-summary
R-CCH#show ip rip database
20.20.0.0/24    auto-summary
20.20.0.0/24    directly connected, GigabitEthernet0/0
20.20.1.0/24    auto-summary
20.20.1.0/24
    [1] via 20.20.4.70, 00:00:04, GigabitEthernet0/2
20.20.2.0/24    auto-summary
20.20.2.0/24
    [1] via 20.20.4.66, 00:00:23, GigabitEthernet0/1
20.20.3.0/25    auto-summary
20.20.3.0/25
    [2] via 20.20.4.70, 00:00:04, GigabitEthernet0/2
20.20.3.128/25  auto-summary
20.20.3.128/25
    [5] via 20.20.4.66, 03:29:03, GigabitEthernet0/1
20.20.4.0/26    auto-summary
20.20.4.0/26
    [5] via 20.20.4.70, 01:11:11, GigabitEthernet0/2
20.20.4.64/30    auto-summary
20.20.4.64/30    directly connected, GigabitEthernet0/1
20.20.4.68/30    auto-summary
20.20.4.68/30    directly connected, GigabitEthernet0/2
20.20.4.72/30    auto-summary
20.20.4.72/30
    [1] via 20.20.4.70, 00:00:04, GigabitEthernet0/2
20.20.4.76/30    auto-summary
20.20.4.76/30
    [2] via 20.20.4.70, 00:00:04, GigabitEthernet0/2
20.20.4.80/30    auto-summary
20.20.4.80/30
    [1] via 20.20.4.66, 00:00:23, GigabitEthernet0/1
20.20.4.84/30    auto-summary
20.20.4.84/30
    [1] via 20.20.4.66, 00:00:23, GigabitEthernet0/1    [1] via 20.20.4.70, 00:00:04,
GigabitEthernet0/2
-

```

C. DHCP

```

R-FDU#show run | section dhcp
ip dhcp excluded-address 20.20.3.1 20.20.3.20
ip dhcp pool FDU_POOL
network 20.20.3.0 255.255.255.128
default-router 20.20.3.1
dns-server 20.20.0.10
domain-name transitline.local
R-FDU#

```

```

R-DPC#show run | section dhcp
ip dhcp excluded-address 20.20.1.1 20.20.1.20
ip dhcp pool DPC_POOL
network 20.20.1.0 255.255.255.0
default-router 20.20.1.1
dns-server 20.20.0.10
domain-name transitline.local
R-DPC#

```

DHCP-PSC server config screenshot

DHCP-PSC

Physical
Config
Services
Desktop
Programming
Attributes

SERVICES

HTTP
DHCP
DHCPv6
TFTP
DNS
SYSLOG
AAA
NTP
EMAIL
FTP
IoT
VM Management
Radius EAP

DHCP

Interface
FastEthernet0
Service
☒ On
☐ Off

Pool Name
serverPool

Default Gateway
20.20.2.1

DNS Server
20.20.0.10

Start IP Address :
20
20
2
100

Subnet Mask:
255
255
255
0

Maximum Number of Users :
156

TFTP Server:
0.0.0.0

WLC Address:
0.0.0.0

Add
Save
Remove

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
serverPool	20.20.2.1	20.20.0.10	20.20.2.100	255.255.2...	156	0.0.0.0	0.0.0.0

PC-DPC-1 , PC-FDU-1 , PC-PSC-1

PC-DPC-1

PhysicalConfigDesktopProgrammingAttributes

IP Configuration

InterfaceFastEthernet0

IP Configuration

☒ DHCP

☐ Static

IPv4 Address

20.20.1.22

Subnet Mask

255.255.255.0

Default Gateway

20.20.1.1

DNS Server

20.20.0.10

IPv6 Configuration

☐ Automatic

☒ Static

IPv6 Address

/

Link Local Address

FE80::2D0:BCFF:FECE:A7A0

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication

MD5

Username

Password

PC-FDU-1

Physical

Config

Desktop

Programming

Attributes

IP Configuration

X

Interface

FastEthernet0

IP Configuration

☒ DHCP

☐ Static

IPv4 Address

20.20.3.22

Subnet Mask

255.255.255.128

Default Gateway

20.20.3.1

DNS Server

20.20.0.10

IPv6 Configuration

☐ Automatic

☒ Static

IPv6 Address

/

Link Local Address

FE80::20B:BEFF:FE09:C623

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication

MD5

Username

Password

PC-PSC-1

Physical Config **Desktop** Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

☒ DHCP ☐ Static

IPv4 Address 20.20.2.101

Subnet Mask 255.255.255.0

Default Gateway 20.20.2.1

DNS Server 20.20.0.10

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address /

Link Local Address FE80::2E0:F7FF:FEB6:3082

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

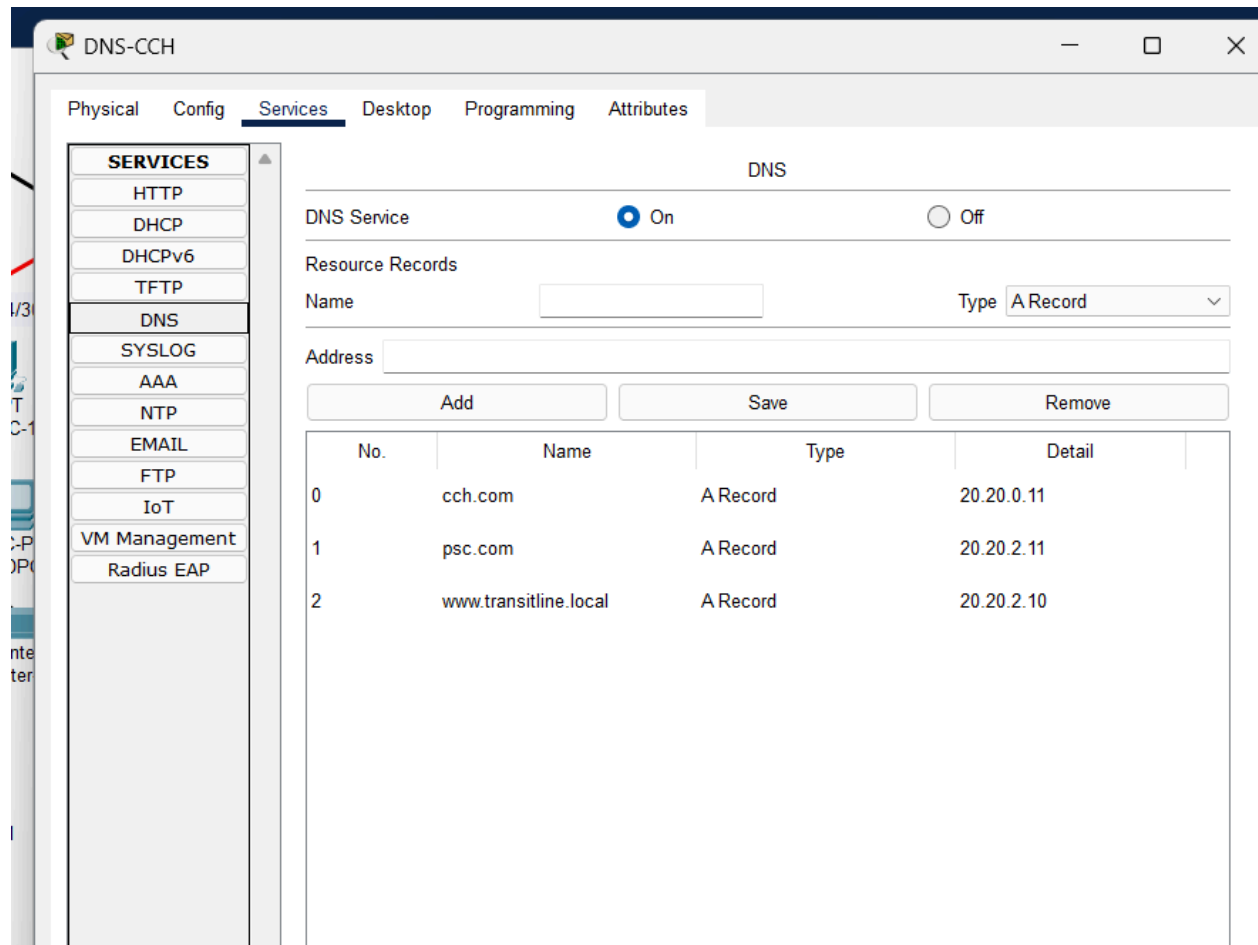
Authentication MD5

Username

Password

D. DNS

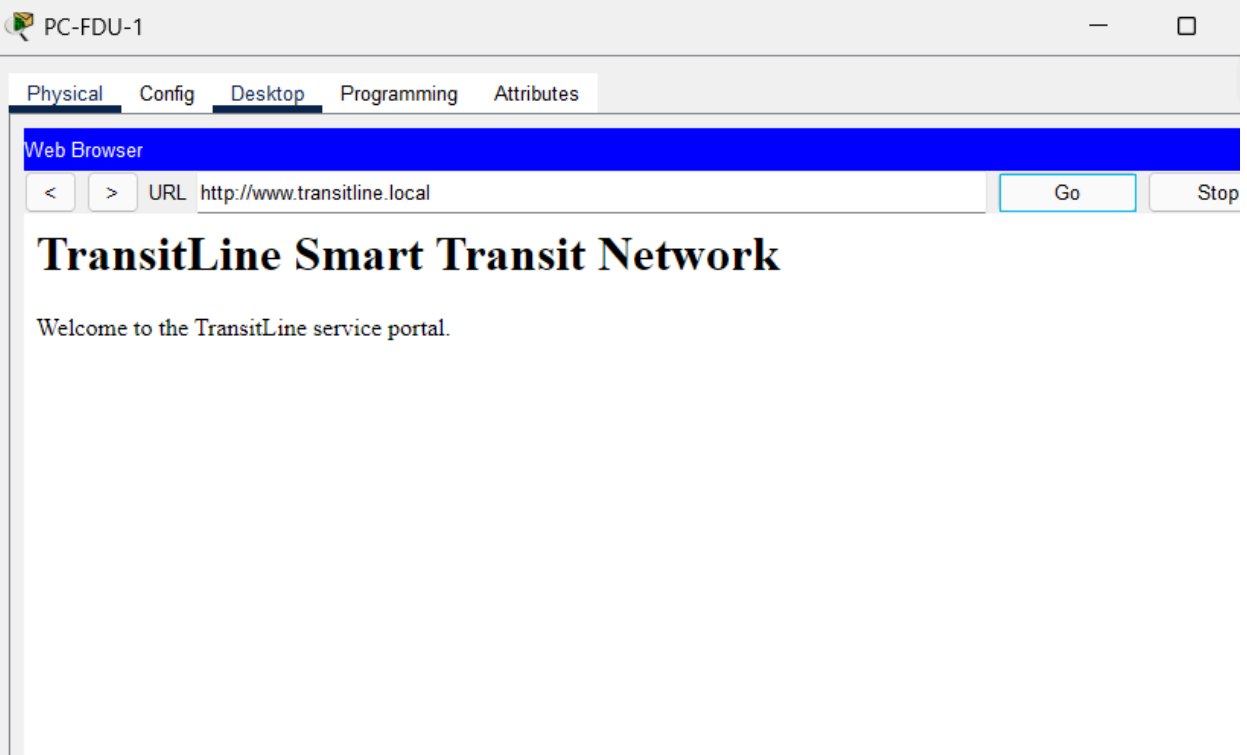
DNS-CCH server config (Services → DNS) — A records

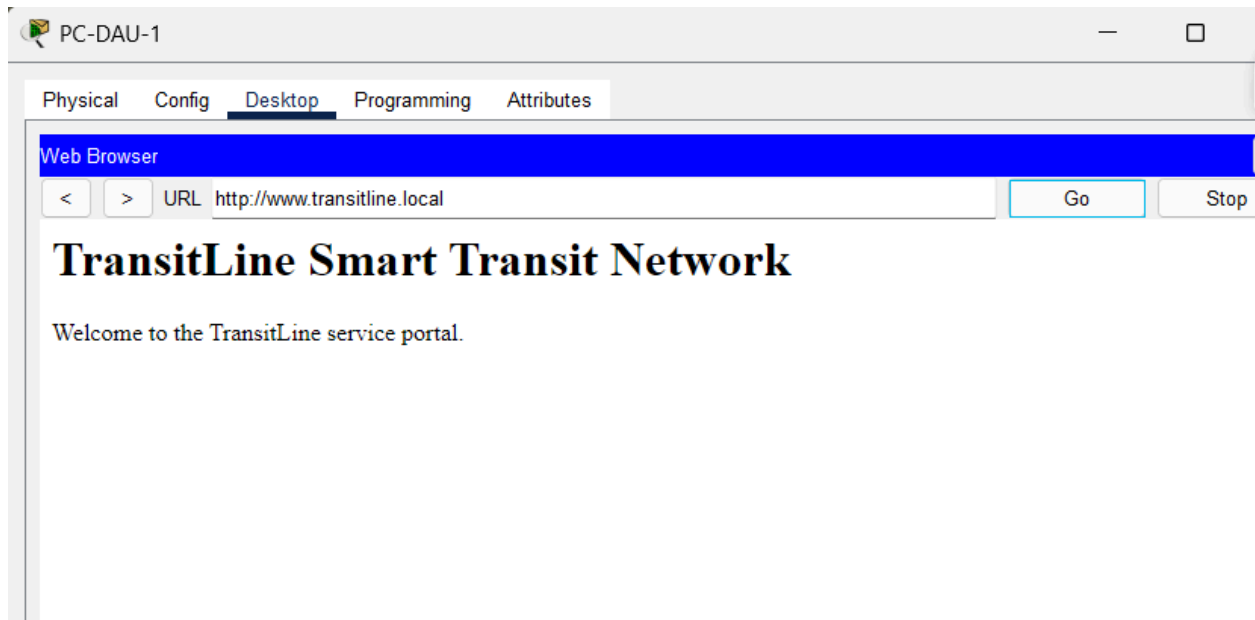
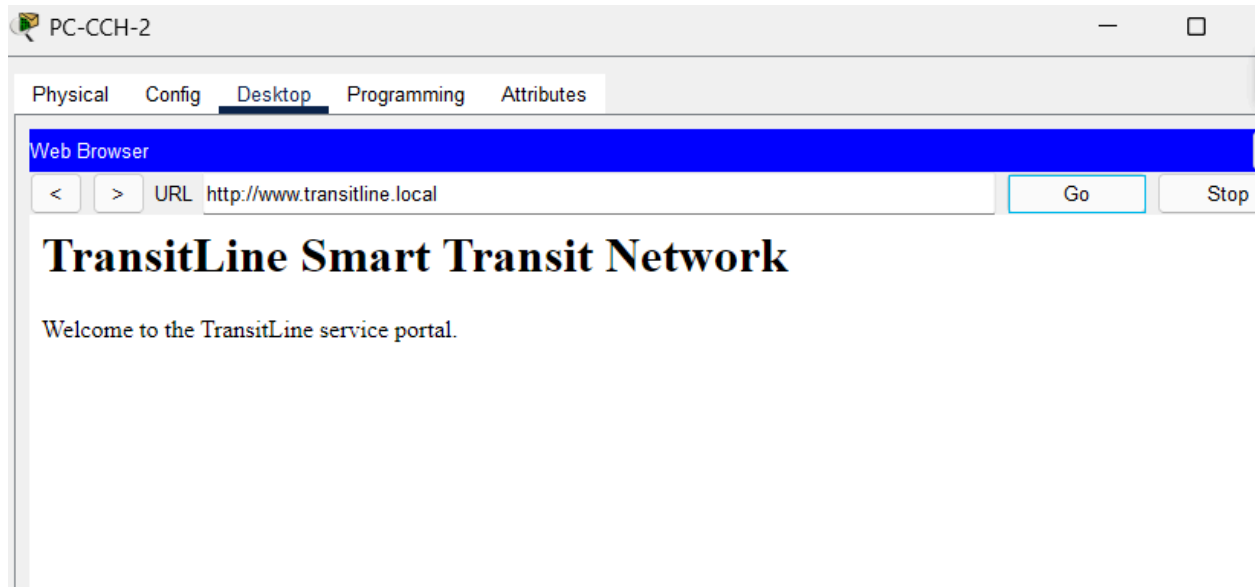


The screenshot shows the DNS-CCH configuration window. The 'Services' tab is selected, and 'DNS' is highlighted in the left sidebar. The 'DNS Service' is turned 'On'. Under 'Resource Records', there are three A records listed in a table.

No.	Name	Type	Detail
0	cch.com	A Record	20.20.0.11
1	psc.com	A Record	20.20.2.11
2	www.transitline.local	A Record	20.20.2.10

E. Web Server





F. Email

EMAIL-CCH and EMAIL-PSC config

The screenshot shows the 'EMAIL-PSC' configuration window with the 'Services' tab selected. The left sidebar lists various services, with 'EMAIL' highlighted. The main area displays the 'EMAIL' configuration, including 'SMTP Service' and 'POP3 Service' both set to 'ON'. The 'Domain Name' is 'psc.com'. The 'User Setup' section shows a list of users: pscuser1, pscuser2, dpcuser1, dpcuser2, fduuser1, fduuser2, rspuser1, rspuser2, dauuser1, and dauuser2. There are buttons for '+', '-', 'Change', and 'Password' on the right.

EMAIL-PSC

Physical Config **Services** Desktop Programming Attributes

SERVICES

- HTTP
- DHCP
- DHCPv6
- TFTP
- DNS
- SYSLOG
- AAA
- NTP
- EMAIL**
- FTP
- IoT
- VM Management
- Radius EAP

EMAIL

SMTP Service ☒ ON ☐ OFF

POP3 Service ☒ ON ☐ OFF

Domain Name:

User Setup

User Password

pscuser1
pscuser2
dpcuser1
dpcuser2
fduuser1
fduuser2
rspuser1
rspuser2
dauuser1
dauuser2

EMAIL-CCH

Physical Config **Services** Desktop Programming Attributes

SERVICES

- HTTP
- DHCP
- DHCPv6
- TFTP
- DNS
- SYSLOG
- AAA
- NTP
- EMAIL**
- FTP
- IoT
- VM Management
- Radius EAP

EMAIL

SMTP Service ☒ ON ☐ OFF

POP3 Service ☒ ON ☐ OFF

Domain Name:

User Setup

User Password

cchuser1
cchuser2

PC mail client config

PC-CCH-2

PhysicalConfigDesktopProgrammingAttributes

Configure MailX

User Information

Your Name:cchuser2

Email Addresscchuser2@cch.com

Server Information

Incoming Mail Server20.20.0.11

Outgoing Mail Server20.20.0.11

Logon Information

User Name:cchuser2

Password:.....

SaveRemoveClearReset

PC-PSC-1

Physical Config **Desktop** Programming Attributes

Configure Mail X

User Information

Your Name: pscuser1

Email Address: pscuser1@psc.com

Server Information

Incoming Mail Server: 20.20.2.11

Outgoing Mail Server: 20.20.2.11

Logon Information

User Name: pscuser1

Password: •••••

Save Remove Clear Reset

Same domain email

Sending mail to cchuser1@cch.com , with subject : test ... Mail
Server: 20.20.0.11
Send Success.

Cancel
Send/Receive

☐ Top

PC-CCH-1

Physical Config **Desktop** Programming Attributes

MAIL BROWSER

X

Mails

Compose

Reply

Receive

Delete

Configure Mail

	From	Subject	Received
1	cchuser2@cch.com	test	Wed Sep 2 2026 04:48:02
2	dauuser1@psc.com	yyyy	Tue Sep 1 2026 01:06:17
3	dpcuser2@psc.com	hi fro me	Tue Sep 1 2026 01:01:54

test
cchuser2@cch.com
Sent : Wed Sep 2 2026 04:48:02

same

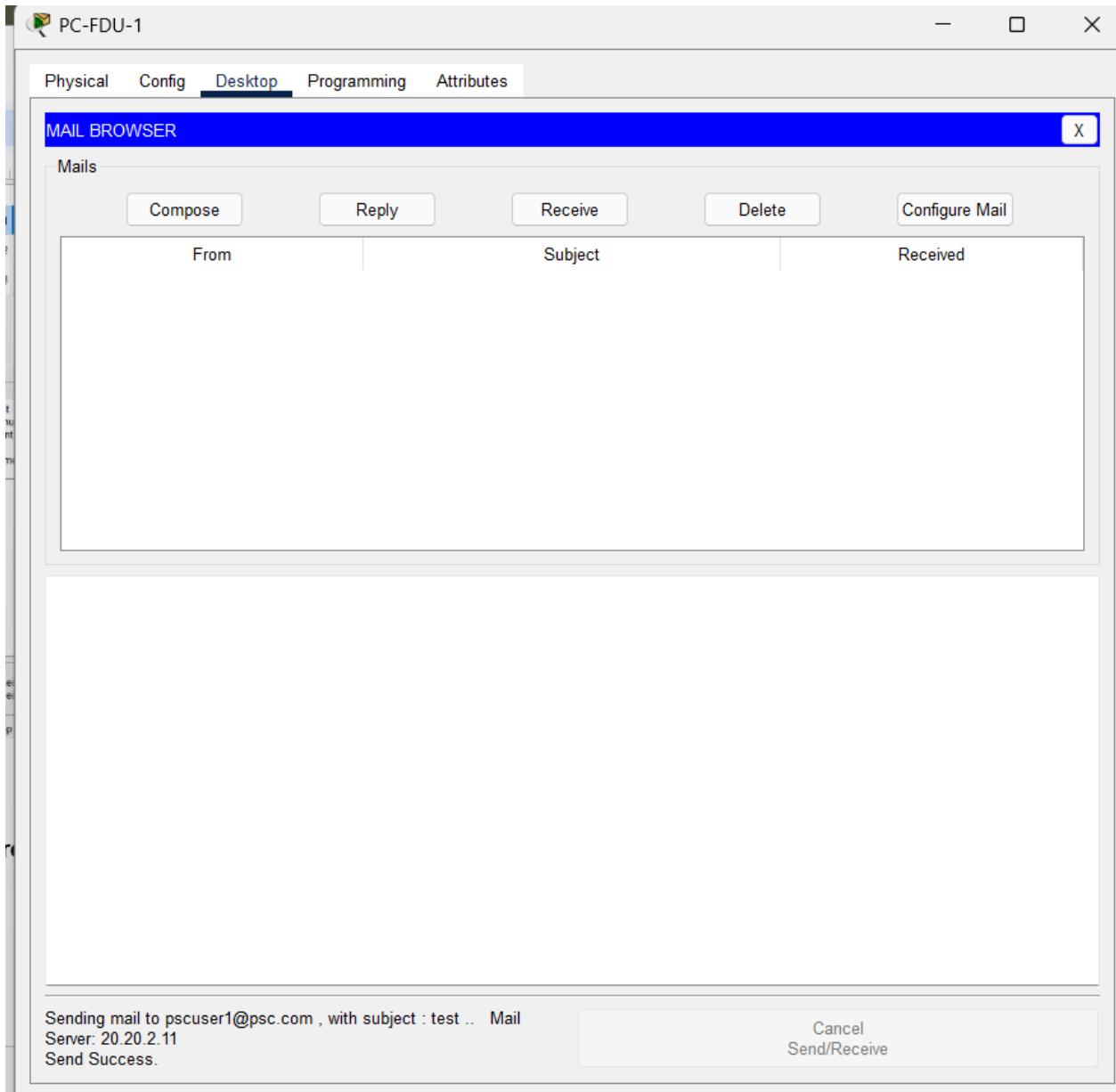
Receiving mail from POP3 Server 20.20.0.11
Receive Mail Success.

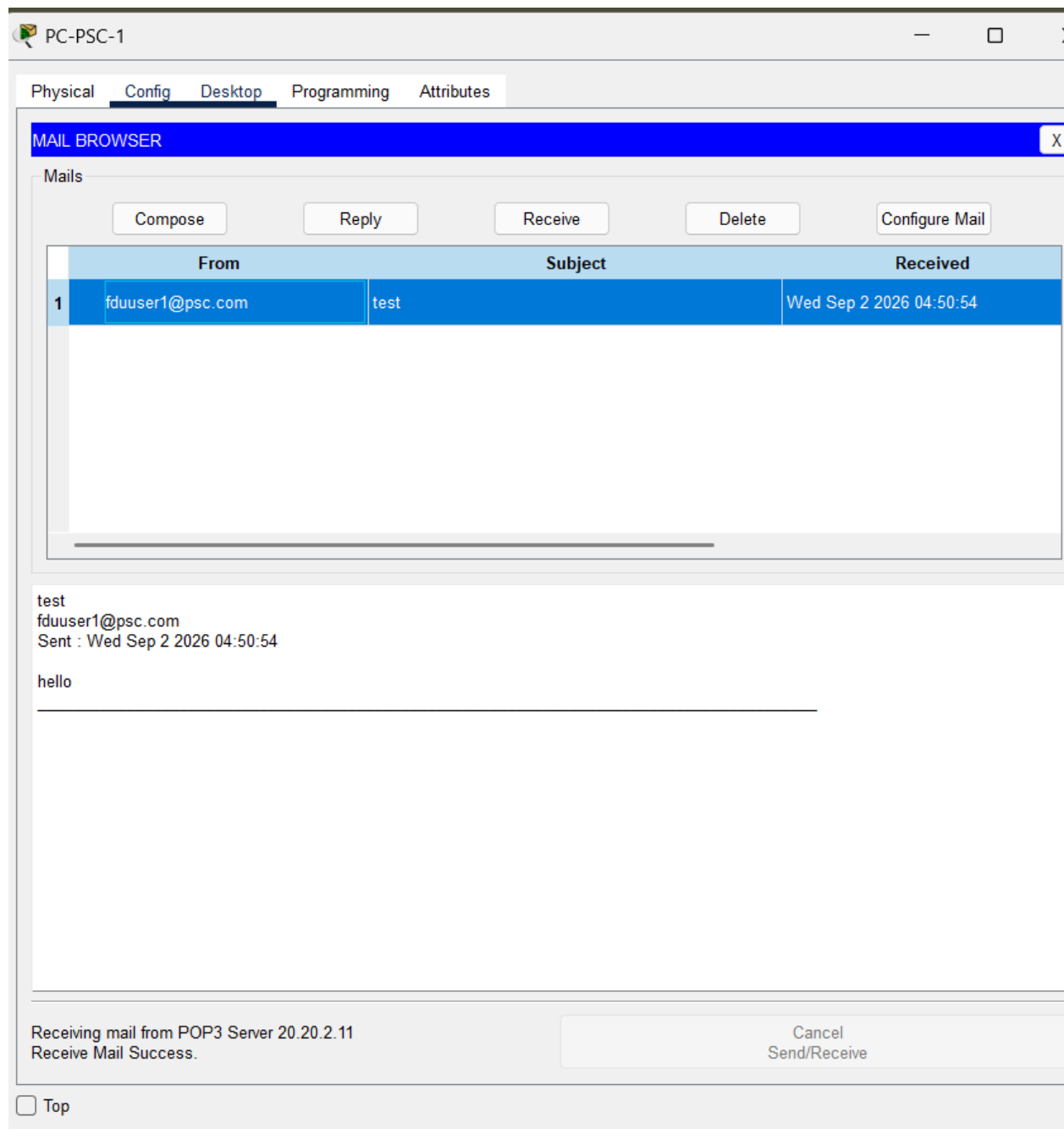
Cancel
Send/Receive

☐ Top

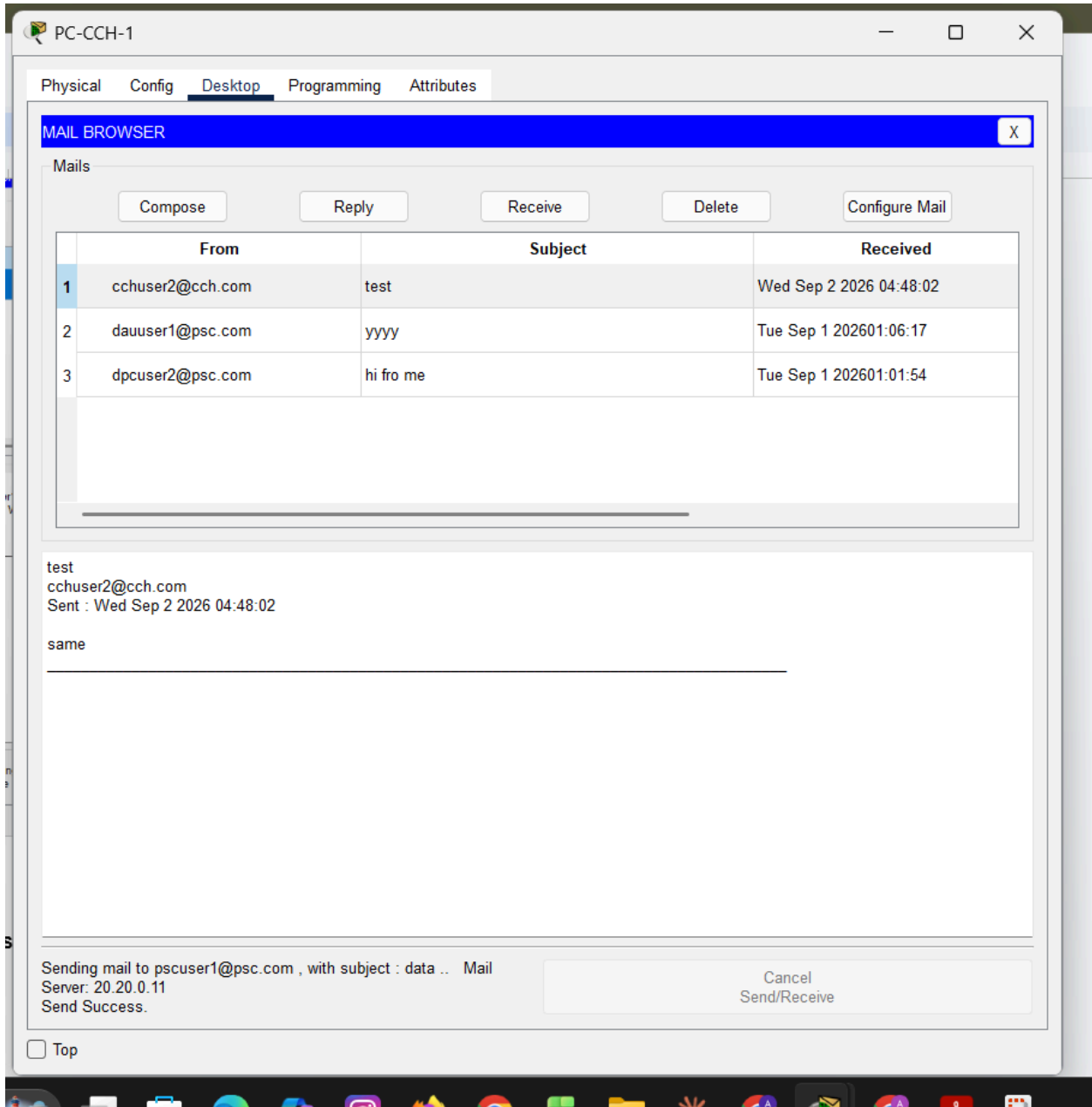
Delete

Different domain email





Cross domain email



PC-PSC-1

Physical

Config

Desktop

Programming

Attributes

MAIL BROWSER

X

Mails

Compose

Reply

Receive

Delete

Configure Mail

	From	Subject	Received
1	cchuser1@cch.com	data	Wed Sep 2 2026 04:51:57
2	fduuser1@psc.com	test	Wed Sep 2 2026 04:50:54

data

cchuser1@cch.com

Sent : Wed Sep 2 2026 04:51:57

hi psc

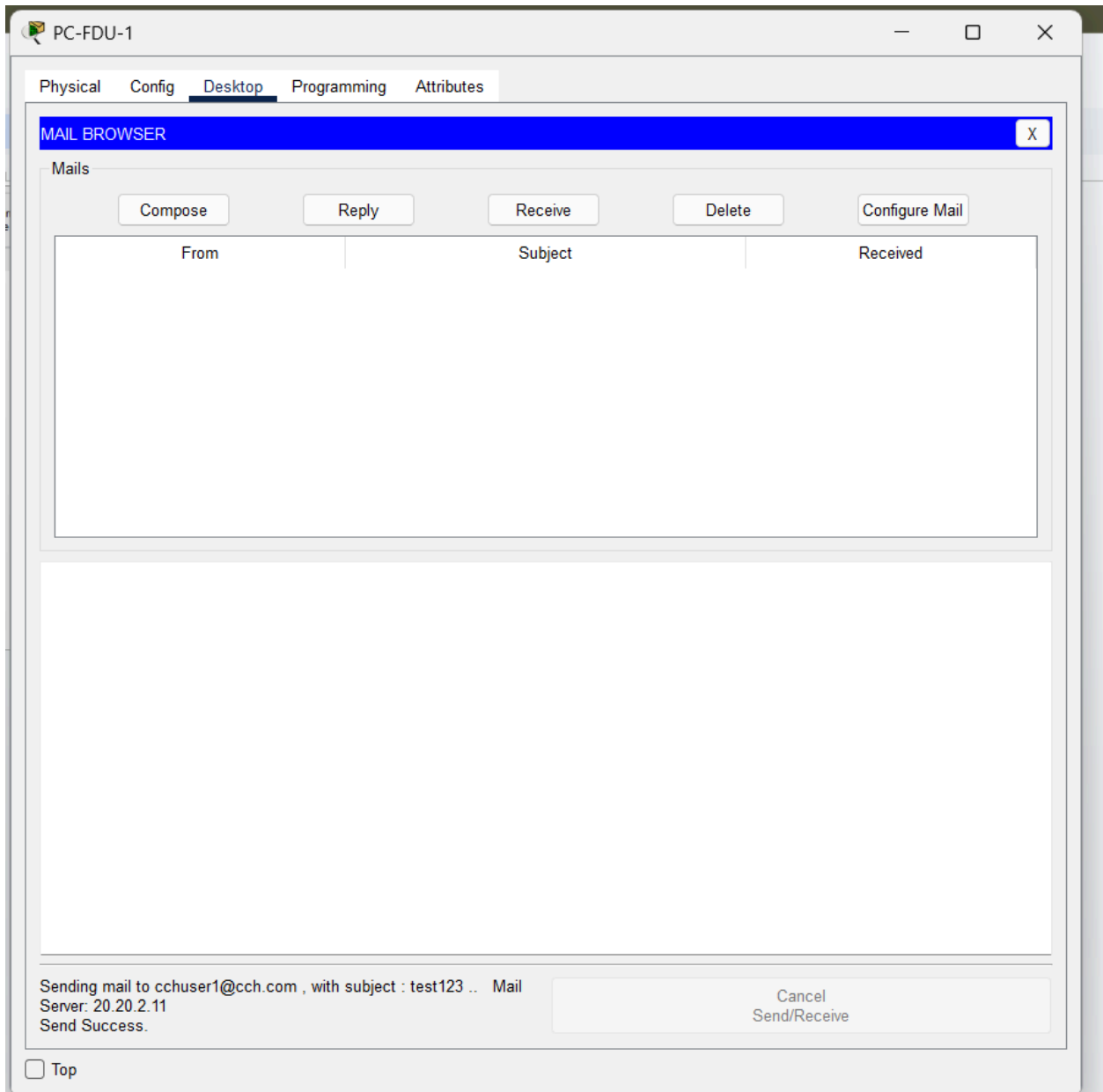
Receiving mail from POP3 Server 20.20.2.11

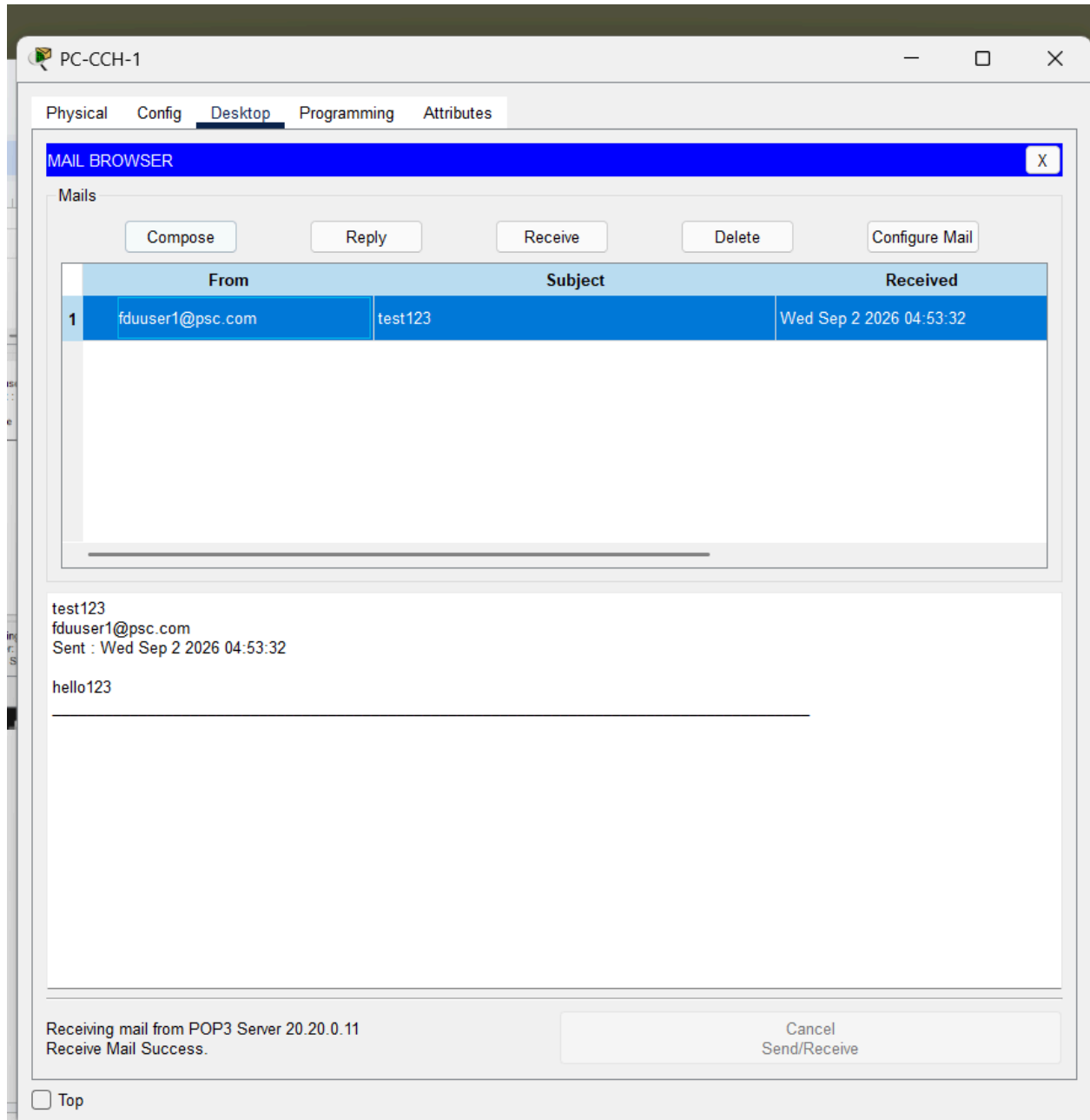
Receive Mail Success.

Cancel

Send/Receive

☐ Top





G. Failover Scenarios

CCH-DPC Link Failure → Floating Static Route

```

R-DPC#show ip route 20.20.2.0
Routing entry for 20.20.2.0/24
Known via "rip", distance 120, metric 2
  Redistributing via rip
  Last update from 20.20.4.69 on GigabitEthernet0/1, 00:00:24 ago
  Routing Descriptor Blocks:
    * 20.20.4.69, from 20.20.4.69, 00:00:24 ago, via GigabitEthernet0/1
      Route metric is 2, traffic share count is 1

```

- ----

R-CCH#configure terminal

R-CCH(config)#interface gigabitEthernet 0/2

R-CCH(config-if)#shutdown

R-CCH(config-if)#end

```

R-DPC#clear ip route *
R-DPC#show ip route 20.20.2.0
Routing entry for 20.20.2.0/24
Known via "static", distance 130, metric 0
  Routing Descriptor Blocks:
    * 20.20.4.86
      Route metric is 0, traffic share count is 1

```

```
C:\>ping 20.20.2.1
```

```
Pinging 20.20.2.1 with 32 bytes of data:
```

```

Reply from 20.20.2.1: bytes=32 time=37ms TTL=254
Reply from 20.20.2.1: bytes=32 time=18ms TTL=254
Reply from 20.20.2.1: bytes=32 time=91ms TTL=254
Reply from 20.20.2.1: bytes=32 time=12ms TTL=254

```

```
Ping statistics for 20.20.2.1:
```

```

    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 12ms, Maximum = 91ms, Average = 39ms

```

***** * * * * * This process on interface GigabitEthernet0/2, changed state

```

R-DPC#clear ip route *
R-DPC#show ip route 20.20.2.0
Routing entry for 20.20.2.0/24
Known via "rip", distance 120, metric 2
  Redistributing via rip
  Last update from 20.20.4.69 on GigabitEthernet0/1, 00:00:01 ago
  Routing Descriptor Blocks:
    * 20.20.4.69, from 20.20.4.69, 00:00:01 ago, via GigabitEthernet0/1
      Route metric is 2, traffic share count is 1

```

- ----

Default Route Check

R-DAU#show ip route | include 0.0.0.0
20.0.0.0/8 is variably subnetted, 14 subnets, 5 masks

R-PSC#show ip route | include 0.0.0.0
20.0.0.0/8 is variably subnetted, 16 subnets, 5 masks

R-CCH#show ip route | include 0.0.0.0
20.0.0.0/8 is variably subnetted, 15 subnets, 5 masks

R-DPC#show ip route | include 0.0.0.0
20.0.0.0/8 is variably subnetted, 16 subnets, 5 masks

R-RSP#show ip route | include 0.0.0.0
Gateway of last resort is 20.20.4.77 to network 0.0.0.0
20.0.0.0/8 is variably subnetted, 5 subnets, 4 masks
S* 0.0.0.0/0 [1/0] via 20.20.4.77

R-FDU#show ip route | include 0.0.0.0
20.0.0.0/8 is variably subnetted, 15 subnets, 5 masks

DAU–PSC Link Failure — Resolved with an Additional Backup Link

The requirement states: *"If the direct link between DAU and PSC fails, DAU must still reach RSP using the recursive static route (DAU → PSC → CCH → DPC → FDU → RSP)."* As originally designed, DAU had only a single physical uplink to the network (via PSC), which meant this exact link could not fail without DAU becoming completely isolated — making the literal scenario undemonstrable within the given topology.

After clarification from the course instructor that additional components may be added where the fixed topology does not otherwise permit a required test, a second physical link was added between R-DAU and R-PSC (subnet 20.20.4.88/30, serial interfaces), purely as a backup path. A floating static route with a higher administrative distance (130) was configured on both R-DAU and R-PSC, so that under normal operation the primary link and its original recursive route remain in full effect, and the backup link stays inactive.

With this addition, the failure scenario was fully and literally demonstrated: the primary DAU–PSC link was shut down, the floating static route automatically activated on both routers, and end-to-end connectivity from DAU to RSP was confirmed via ping (0% loss) and traceroute, with the path passing through the backup link to PSC and then through CCH → DPC → FDU exactly as required. The link was then restored and the primary route resumed automatically, confirming correct failover and recovery behaviour.

```
R-DAU(config)#end
R-DAU#
%SYS-5-CONFIG_I: Configured from console by console

R-DAU#show run | include ip route
ip route 20.20.4.0 255.255.255.192 20.20.4.81
ip route 20.20.0.0 255.255.255.0 20.20.4.81
ip route 20.20.1.0 255.255.255.0 20.20.4.81
ip route 20.20.2.0 255.255.255.0 20.20.4.81
ip route 20.20.3.0 255.255.255.128 20.20.4.81
ip route 20.20.4.64 255.255.255.252 20.20.4.81
ip route 20.20.4.68 255.255.255.252 20.20.4.81
ip route 20.20.4.72 255.255.255.252 20.20.4.81
ip route 20.20.4.76 255.255.255.252 20.20.4.81
ip route 20.20.4.84 255.255.255.252 20.20.4.81
ip route 20.20.4.0 255.255.255.192 20.20.4.90 130
R-DAU#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
R-DAU#show ip route 20.20.4.0
Routing entry for 20.20.4.0/26
Known via "static", distance 1, metric 0
  Routing Descriptor Blocks:
    * 20.20.4.81
      Route metric is 0, traffic share count is 1

R-DAU#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R-DAU(config)#interface gigabitEthernet 0/1
R-DAU(config-if)#shutdown

R-DAU(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to administratively down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to down

R-DAU(config-if)#end
R-DAU#
%SYS-5-CONFIG_I: Configured from console by console

R-DAU#
-----

R-DAU#clear ip route *
R-DAU#show ip route 20.20.4.0
Routing entry for 20.20.4.0/26
Known via "static", distance 130, metric 0
  Routing Descriptor Blocks:
    * 20.20.4.90
      Route metric is 0, traffic share count is 1

R-DAU#show ip route 20.20.4.0
Routing entry for 20.20.4.0/26
Known via "static", distance 130, metric 0
  Routing Descriptor Blocks:
    * 20.20.4.90
      Route metric is 0, traffic share count is 1

R-DAU#
```

```

Pinging 20.20.4.1 with 32 bytes of data:

Reply from 20.20.4.1: bytes=32 time=20ms TTL=250
Reply from 20.20.4.1: bytes=32 time=2ms TTL=250
Reply from 20.20.4.1: bytes=32 time=28ms TTL=250
Reply from 20.20.4.1: bytes=32 time=9ms TTL=250
|
Ping statistics for 20.20.4.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 28ms, Average = 14ms

C:\>tracert 20.20.4.1

Tracing route to 20.20.4.1 over a maximum of 30 hops:

  1  0 ms      0 ms      0 ms      20.20.3.129
  2  7 ms      2 ms      6 ms      20.20.4.90
  3  2 ms      0 ms      0 ms      20.20.4.65
  4  1 ms      5 ms      2 ms      20.20.4.70
  5  6 ms      1 ms      0 ms      20.20.4.74
  6  1 ms      1 ms      0 ms      20.20.4.1

Trace complete.

```

2015-0-CONFIG_1. Configured from console by console

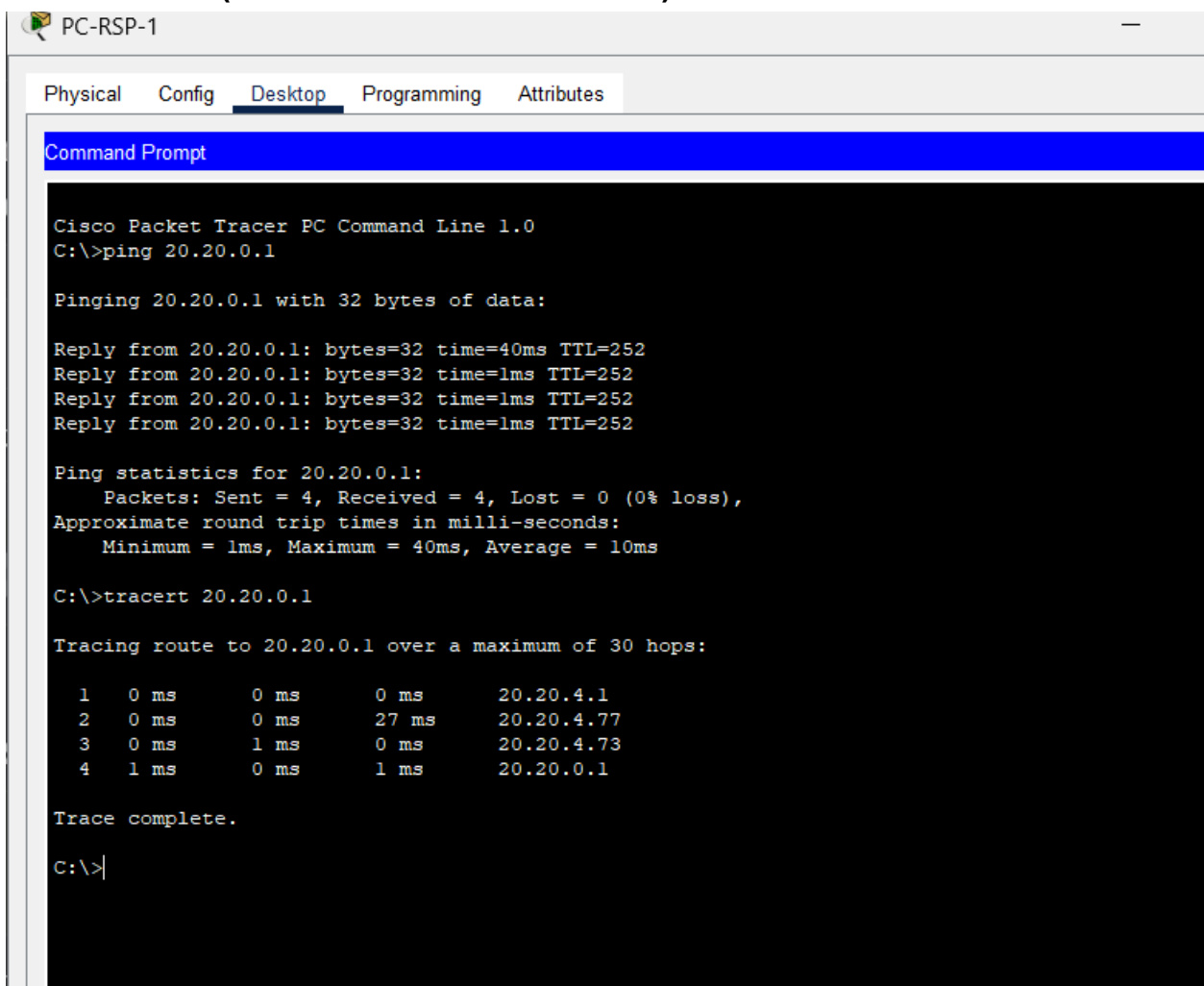
```

R-PSC#clear ip route *
R-PSC#show ip route 20.20.3.128
Routing entry for 20.20.3.128/25
Known via "static", distance 130, metric 0
  Routing Descriptor Blocks:
    * 20.20.4.89
      Route metric is 0, traffic share count is 1

```

General Connectivity — Multi-hop Route Proof

RSP → CCH (Exit-interface static route)



```
PC-RSP-1
Physical Config Desktop Programming Attributes
Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 20.20.0.1

Pinging 20.20.0.1 with 32 bytes of data:

Reply from 20.20.0.1: bytes=32 time=40ms TTL=252
Reply from 20.20.0.1: bytes=32 time=1ms TTL=252
Reply from 20.20.0.1: bytes=32 time=1ms TTL=252
Reply from 20.20.0.1: bytes=32 time=1ms TTL=252

Ping statistics for 20.20.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 40ms, Average = 10ms

C:\>tracert 20.20.0.1

Tracing route to 20.20.0.1 over a maximum of 30 hops:

  1  0 ms    0 ms    0 ms    20.20.4.1
  2  0 ms    0 ms   27 ms    20.20.4.77
  3  0 ms    1 ms    0 ms    20.20.4.73
  4  1 ms    0 ms    1 ms    20.20.0.1

Trace complete.

C:\>
```

DAU → RSP (Recursive static route)

```
C:\>ping 20.20.4.1

Pinging 20.20.4.1 with 32 bytes of data:

Reply from 20.20.4.1: bytes=32 time=1ms TTL=250
Reply from 20.20.4.1: bytes=32 time=1ms TTL=250
Reply from 20.20.4.1: bytes=32 time=1ms TTL=250
Reply from 20.20.4.1: bytes=32 time=11ms TTL=250

Ping statistics for 20.20.4.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 11ms, Average = 3ms

C:\>tracert 20.20.4.1

Tracing route to 20.20.4.1 over a maximum of 30 hops:

  1  0 ms      0 ms      0 ms      20.20.3.129
  2  0 ms      0 ms      0 ms      20.20.4.81
  3  0 ms      0 ms      0 ms      20.20.4.65
  4  0 ms      0 ms      0 ms      20.20.4.70
  5  0 ms      0 ms      0 ms      20.20.4.74
  6  0 ms      0 ms      0 ms      20.20.4.1

Trace complete.
```